

Problem 7

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Problem 1 Given the expression for $f(x)$, find the derivative of f^{-1} at the given point on the graph of f^{-1} , without solving for f^{-1} .

- (a) $f(x) = x^2 + 1$ (for $x \geq 0$); the point on the graph of f^{-1} : $(5, 2)$.

Verify your answer by evaluating the derivative of f^{-1} at the given point.

Explanation. $(f^{-1})'(5) = \frac{1}{f'(2)}$. Since $f'(x) = 2x$, $f'(2) = 4$. Thus,

$$(f^{-1})'(5) = \frac{1}{4}.$$

$$\text{Verifying: } f^{-1}(x) = \sqrt{x-1} \implies \frac{d}{dx} f^{-1}(x) = \frac{1}{2\sqrt{x-1}} \implies \left[\frac{d}{dx} f^{-1} \right]_{x=5} = \frac{1}{2\sqrt{5-1}} = \frac{1}{4}$$

- (b) $f(x) = x^2 - 2x - 3$ (for $x \leq 1$); the point on the graph of f^{-1} : $(12, -3)$.

Explanation. $(f^{-1})'(12) = \frac{1}{f'(-3)}$. Since $f'(x) = 2x - 2$, $f'(-3) = -6 - 2 = -8$. Thus, $(f^{-1})'(12) = -\frac{1}{8}$.

Notice that for this problem, finding the inverse of $f(x)$ involves complicated algebra. In this case, it's much easier to find the derivative of f^{-1} without solving for f^{-1} .